Supplementary material

Procedure Selection versus Workflow Structuring in Local RAG for Institutional SOP Question Answering: An Ablation Study

Table S1: Complete evaluation results across all 10 systems on 40 verified benchmark questions, reporting mean [95% bootstrap confidence intervals].

System	KFR ↑	Coverage ↑	TIR ↓	AMR ↓	Refusal acc. ↑	False ref. ↓	Prompt tok.
Naive RAG	0.712 [0.571, 0.840]	0.727 [0.455, 1.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.600 [0.200, 1.000]	0.000 [0.000, 0.000]	916
GraphRAG (entity)	0.698 [0.560, 0.829]	0.687 [0.415, 0.920]	0.083 [0.000, 0.250]	0.000 [0.000, 0.000]	0.600 [0.200, 1.000]	0.000 [0.000, 0.000]	949
Workflow GraphRAG	0.955 [0.917, 0.990]	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	1525
Selection-Matched Text RAG (ablation)	0.962 [0.914, 1.000]	0.975 [0.926, 1.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	1479
Hybrid RAG	0.683 [0.536, 0.819]	0.712 [0.439, 0.970]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.600 [0.200, 1.000]	0.000 [0.000, 0.000]	882
Agentic RAG	0.712 [0.571, 0.840]	0.702 [0.430, 0.950]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.600 [0.200, 1.000]	0.000 [0.000, 0.000]	1815
Corrective RAG	0.569 [0.414, 0.719]	0.636 [0.364, 0.909]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	1.000 [1.000, 1.000]	0.371 [0.229, 0.543]	725
Multimodal-layout RAG	0.702 [0.562, 0.831]	0.702 [0.430, 0.950]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.600 [0.200, 1.000]	0.000 [0.000, 0.000]	994
LLM only	0.113 [0.050, 0.183]	0.105 [0.000, 0.209]	0.333 [0.000, 0.667]	0.444 [0.000, 1.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	107
Full context	0.962 [0.914, 1.000]	0.944 [0.871, 1.000]	0.027 [0.000, 0.082]	0.000 [0.000, 0.000]	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]	2722

Table S2: The 17 hand-curated relational triples of the custom entity-graph GraphRAG baseline, as defined in `src/engine.py`. Entities and relations are kept in their original Indonesian form.

#	Head entity	Tail entity	Relation
1	cuti akademik	izin cuti	prosedur pengajuan
2	cuti akademik	aktif kembali	alur lanjutan
3	cuti akademik	ketua program studi	rekomendasi & persetujuan
4	cuti akademik	dekan	penerbitan SK izin
5	cuti akademik	spp/ukt	syarat bebas tunggakan
6	legalisir	ijazah	dokumen objek
7	legalisir	transkrip	dokumen objek
8	legalisir	subbagian akademik	verifikasi & validasi
9	beasiswa	rekomendasi beasiswa	dokumen pengajuan
10	beasiswa	wakil dekan i	pengesahan surat
11	ukt	keterlambatan pembayaran	permohonan dispensasi
12	ukt	bagian keuangan fsm	verifikasi pembayaran
13	irs	pengisian irs	registrasi akademik per semester
14	irs	dosen wali	bimbingan & persetujuan online
15	irs	siap undip	portal sistem informasi
16	organisasi mahasiswa	proposal kegiatan	pengajuan persetujuan
17	organisasi mahasiswa	wakil dekan i	persetujuan kegiatan

Table S3: One benchmark question per category from `sop_benchmark_v1.json`, with English glosses.

ID	Category	Question (Indonesian) / English gloss
CUTI-P1	Procedural	*Apa saja tahapan mengajukan cuti akademik di FSM dari awal sampai surat izin keluar, dan siapa yang mengerjakan tiap tahap?* What are the steps for applying for academic leave at FSM, from the start until the permit letter is issued, and who performs each step?
CUTI-A1	Actor	*Setelah form cuti saya ditandatangani Kaprodi, form itu diserahkan ke siapa dan diproses oleh bagian apa?* After the Head of Study Programme signs my leave form, to whom is it submitted and which unit processes it?
CUTI-D1	Documents	*Dokumen apa saja yang harus dilampirkan saat mengajukan cuti akademik?* Which documents must be attached when applying for academic leave?
CUTI-T1	Time/location	*Berapa lama maksimal proses izin cuti akademik, dan di mana saya mengambil Surat Izin Dekan?* What is the maximum processing time for an academic leave permit, and where do I collect the Dean's permit letter?
X-1	Cross-SOP	*Apa perbedaan dokumen yang dibawa untuk mengajukan cuti akademik dengan untuk aktif kembali setelah cuti?* How do the documents required for academic leave differ from those required for reactivation after leave?
NA-1	Unanswerable	*Berapa biaya legalisir ijazah per lembar di FSM?* How much does diploma legalisation cost per sheet at FSM?